A Theory of Economic Slack

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Preface

This book is based for the most part on research that I conducted during graduate school at the University of California–Berkeley under the supervision of George Akerlof and Yuriy Gorodnichenko (2006–2010), and for fifteen years after that in collaboration with Emmanuel Saez (2010–2025).

In this preface, I'd like to explain what motivated the research, and to provide the key references on which the book is based. The purpose is not to describe results, but to explain how the theory presented in the book emerged from a sequence of research questions, themselves motivated by macroeconomic puzzles and policy debates.

Why do job seekers queue in recessions?

Early on in graduate school I read about Okun's misery index—the sum of the unemployment rate and inflation rate, designed as a simple measure of welfare in the US economy.[1] A low index means that the economy is doing well; a high index that it is doing poorly. The misery index posits that inflation and unemployment are the two main diseases affecting developed economies. The discovery of the misery index prompted me to start working on inflation. I particularly wanted to understand why seemingly tiny changes in the price of a baguette in France triggered such outrage from customers. But that project proved to be too challenging, so I turned to the other component of the misery index: unemployment.

A first book that George advised me to read to start thinking about unemployment and business cycles was *Prices & Quantities* by Arthur Okun (1981). In it, Okun sketches a new macroeconomic framework in which labor and product markets are organized around implicit contracts between sellers and buyers. Although the slackish business cycle model developed in the book is quite different from the framework that Okun had in mind, the book draws in numerous places on evidence that Okun collected, observations that he made, and references that he provided. Generally, the book shares Okun's conviction that

[1] See Asher, Defina, and Thanawala (1993) for a history of the misery index.

Walrasian theory is inappropriate to describe labor and product markets, and that prices and wages are determined by norms that are somewhat insulated from cyclical forces.

George also suggested that I read *Equilibrium Unemployment Theory* by Christopher Pissarides (2000)—the leading textbook on unemployment theory. The textbook introduces the Diamond-Mortensen-Pissarides (DMP) model of the labor market, which was developed by Pissarides together with Peter Diamond and Dale Mortensen in the 1970s and 1980s. For their work, they received the Nobel Prize in Economics in 2010.[2]

An idea in the textbook that stuck with me is that it is possible, and in fact quite fruitful, to summarize the complex matching process on the labor market by a matching function—without modeling the matching process explicitly.[3] The idea plays a central role in this book, as trade in all markets is mediated by a well-behaved matching function. The function gives the number of trades that actually take place when a certain number of buyers and sellers are looking for each other on a market.

Reading the textbook, however, I was struck that the DMP model does not allow for the possibility—which seemed relevant in real life—that sometimes the economy just does not have enough jobs for all workers. This is an old, very Keynesian idea. Keynes had seen what was happening at the time of the Great Depression; he had seen the queues of workers waiting to apply for jobs, or to be hired, in front of job bureaus and factory gates. The photos collected in this preface, taken across the United States by Dorothea Lange, illustrate what Keynes (1936) had seen and wrote about in the *General Theory*: that in bad times there are just not enough jobs for everyone at the going wage, that jobs are rationed.

By ignoring job rationing, I felt, the DMP model cannot provide a good description of what happens in the labor market. In bad times, in the real world, workers queue for jobs. In that situation, a factory manager can just pick any worker that they want immediately; it is costless for firms to hire; matching frictions are irrelevant. And yet, workers visibly queue for jobs. So it seems that something else happens in this situation: jobs are lacking.[4]

Thus, I became convinced that to describe the labor market better, we must allow for job rationing. My PhD thesis therefore developed a labor market model that features not only frictional unemployment—as in the DMP model—but also rationing unemployment—as in the *General Theory*. In the model, firms might not want to create enough jobs for all workers who want to work. The main result is that in recessions, frictional unemployment

[2] Albrecht (2011) reviews the early work by Diamond, Mortensen, and Pissarides, and the literature that built on it. Diamond (2011), Mortensen (2011), and Pissarides (2011) provide their own perspective on their research and the literature in their Nobel lectures.

[3] This idea goes back to work by Pissarides (1979, 1984, 1985, 1986). Before that, people usually modeled specific trading processes that resulted in trading probabilities between 0 and 1, often from the urn-ball model that is commonly used in probability.

[4] In the 1970s and 1980s, the literature on efficiency wages built models producing job rationing. But the models fell out of favor, maybe because they could not easily be integrated into the fast-growing DMP framework. For surveys of the efficiency-wage literature, see Akerlof (1984), Yellen (1984), and Katz (1986). See Akerlof and Yellen (1990) for one of the most mature models in that literature.

A. Howard Street, known as "Skid Row," the district of the unemployed (San Francisco, CA, 1937)

B. Unemployed men sitting on the sunny side of the San Francisco Public Library (San Francisco, CA, 1937)

C. Part of the daily lineup outside the State Employment Service Office (Memphis, TN, 1938)

D. Typical scene reflecting large population of unemployed in desperate need of work and looking for jobs (Camden, NJ, 1935)

Job rationing during the Great Depression in the United States

Sources. A: Dorothea Lange, available at `https://www.loc.gov/pictures/item/2017769707/`. B: Dorothea Lange, available at `https://www.loc.gov/pictures/item/2017769687/`. C: Dorothea Lange, available at `https://lccn.loc.gov/2017770574`. D: Franklin D. Roosevelt Library Public Domain Photographs, available at `https://catalog.archives.gov/id/195658`.

is small and most of unemployment comes from job rationing. This core idea was then published in the *American Economic Review* in 2012 (Michaillat 2012).

The AER paper introduces one of the book's key ideas: that a matching structure allows us to bridge the gap between older, Keynesian theories that emphasize the role of demand in macroeconomics and modern theories that emphasize the role of supply. In doing so, the book offers a theory that simultaneously describes the macroeconomic effects of demand-side and supply-side factors. This idea is found throughout the book, but a streamlined version of the AER labor market model is presented in chapter 10.

A second idea in the paper, which is also an important part of the book, is that matching models are well suited to accommodate realistic, somewhat rigid price and wage norms, and that such price and wage rigidity generates realistic business cycle fluctuations. That idea was developed by Hall (2005), which showed that fixed wages are compatible with bilateral efficiency in matching models, and that with fixed wages labor market matching models generate realistic labor market fluctuations. A further insight from Blanchard and Gali (2010) is that in fact, wages do not have to be fixed: they can simply be a rigid function of productivity. In the real world, very few prices are determined at auction, as the Walrasian model assumes. Instead, most sellers and buyers are engaged in repeated, long-term relationships where norms of fairness matter a great deal. In a paper with Erik Eyster and Kristof Madarasz, that stemmed from the baguette project, but greatly developed and improved it, we document these norms and model pricing under fairness concerns (Eyster, Madarasz, and Michaillat 2021). The book does not use the pricing model developed in that paper, but it uses several of the facts documented there to build realistic price norms, especially in chapters 5 and 6.

Towards the end of my PhD thesis, I stumbled on an interesting property of the model: it is highly state dependent, in that it operates very differently in booms and slumps. As a result, some policies are systematically more effective when the economy is slack, while other policies are systematically more effective when the economy is tight. For instance, the fiscal multiplier is larger when unemployment is high, thus explaining Yuriy and Alan Auerbach's discovery that fiscal multipliers are higher during recessions (Auerbach and Gorodnichenko 2012). This result was published in the *American Economic Journal: Macroeconomics* in 2014 (Michaillat 2014). The state dependence is present throughout the book in various contexts, but especially in chapter 8, where the effects of shocks are strongly slack dependent, and in chapters 11 and 13, where we obtain slack-dependent policy multipliers.

Should unemployment insurance be extended in recessions?

I went on the job market in 2009–2010, in the aftermath of the Great Recession. One of the big policy questions at the time was whether to make unemployment insurance more generous. Emmanuel attended a mock job talk that I gave in December 2009, thought that the model that I had developed in my thesis could be helpful to think about the question, and reached out to see if I wanted to collaborate with him and Camille Landais, who was doing a postdoc at Berkeley at the time, on a project to determine how unemployment insurance should be adjusted over the business cycle.

The reason why the model was promising is that public finance only took into account supply-side factors—how unemployment insurance affected job search—but did not have the tools to include demand-side factors—the number of workers that firms are willing to hire. The key idea was that if firms do not want to hire more workers, then searching more for jobs will only result in a giant rat race, but not additional employment, which would severely reduce the purported social cost of higher unemployment insurance and make an unemployment insurance extension in bad times more desirable.

So in the summer of 2010, Camille, Emmanuel, and I started working on the following question: how should the generosity of unemployment insurance respond to unemployment fluctuations? We found that in the presence of job rationing, optimal unemployment insurance is more generous than the conventional Baily (1978)-Chetty (2006) level when the labor market is inefficiently slack and below it when the labor market is inefficiently tight. We also show how to identify empirically the presence of job rationing: there is job rationing if and only if the macro effect of unemployment insurance on employment is less than its micro effect. We collect evidence of job rationing in the US labor market. The implication is that unemployment insurance should be more generous in bad times than in good times—as it is in practice in the United States. The project was ultimately published in 2018, in a pair of papers in the *American Economic Journal: Economic Policy* (Landais, Michaillat, and Saez 2018b,a). This analysis of unemployment insurance is discussed in chapter 11.

The AEJ papers aimed to express the formula for optimal unemployment insurance in sufficient statistics. The idea was to describe optimal policy with formulas expressed in terms of statistics that apply across a range of models (not just a specific model) and that are directly measurable (unlike model parameters).[5] Sufficient statistics make the policy tradeoffs transparent and rooted in empirical evidence, but they were not used to study macro, market-level effects of policy. Market-level effects of policies were typically studied in clunky, structural models that kept the policy tradeoffs and mechanisms obscure. The AEJ papers brought sufficient statistics to the macrodesign of policy; the book follows this methodology, especially in chapter 9 and in part V.

[5] See Chetty (2009) and Kleven (2021) for surveys. Emmanuel's job-market paper played a central role in the shift towards sufficient statistics in public economics (Saez 2001).

How are aggregate demand and labor demand connected?

Despite the difficulties encountered in our project on unemployment insurance, Emmanuel and I decided to keep working together with the framework. A natural question that arose is why labor demand is sometimes high and sometimes low. In the labor market model from my thesis, which we then used in our unemployment insurance project, fluctuations in labor productivity lead to fluctuations in labor demand. This is the same as in the DMP model. But in reality, labor productivity is unlikely to be the main driver of business cycles. Surely aggregate demand fluctuations, stemming from changes in monetary policy or animal spirits, are bound to influence labor demand. The question is therefore how to connect aggregate demand to labor demand.

The answer, it turns out, is slack on the product market. Unemployment is the most studied form of slack but slack also exists on the product market: firms are always looking for customers to buy their products and services. Using the architecture developed by Barro and Grossman (1971), we build a model with slack on both the product market and the labor market. The resulting model blends three components of unemployment: a Keynesian component because unemployment depends on how difficult it is for firms to sell their goods; a classical component because unemployment depends on the real wage; and a frictional component because unemployment depends on how costly it is for firms to recruit workers.[6] This work was published in the *Quarterly Journal of Economics* in 2015 (Michaillat and Saez 2015) and is covered in chapter 15.

While our QJE paper offers a simple model that is useful to understand why slack exists and how slack on different markets interact, it is static so cannot be used to think about interest rates, which inherently involve dynamic savings decisions. And interest rates are the tool that central banks use to stabilize the economy. So, Emmanuel and I next developed a dynamic model and inserted interest rates into it. In this monetary model, unemployment is determined by the intersection of an aggregate-demand curve, stemming from households' consumption-saving decisions, and an aggregate-supply curve, corresponding to the Beveridge curve. Monetary policy influences the real interest rate, which is a key determinant of the aggregate-demand curve, so monetary policy can be used to stabilize the economy. This work was published in the *Oxford Economic Papers* in 2022 (Michaillat and Saez 2022). The model is covered in chapter 13 and the implications for monetary policy are covered in chapter 16.

What happens during long zero-lower-bound episodes?

One of the challenges that we faced in developing the OEP paper was how to build a nondegenerate aggregate demand curve. By nondegenerate I mean an aggregate demand

[6]This typology comes from Malinvaud (1977, figure 3).

curve that is downward-sloping, just like the old IS curve, and that can be used to study a monetary model at the zero lower bound or away from it, without requiring any specific monetary rules. The standard aggregate demand, arising from a standard Euler equation, actually fails these requirements, because it is horizontal: it only tolerates a real interest rate equal to the time discount factor.

We discovered that it was possible to obtain a nondegenerate aggregate demand curve by assuming that people had wealth in their utility function. The justification for the assumption is that wealth is a marker of social status, and people value high social status. Thanks to this assumption, the model behave well at the zero lower bound, even if the zero-lower-bound episode lasted a very long time, just like in the aftermath of the Great Recession in the United States.

The assumption of wealth in the utility function is extremely useful to produce a well-behaved aggregate demand curve, so it is used throughout part IV and part V. That assumption had been used sporadically in various macroeconomic contexts since Kurz (1968) first introduced it, especially to understand saving and investment behavior, and relatedly, long-run growth.[7] But it had not been used in business cycle research, especially in combination with a slackish model, so the paper, written in 2014, was not published before 2022.

What we did manage to publish was a related paper showing that the wealth-in-the-utility assumption resolved all the anomalies of the New Keynesian model at the zero lower bound. The paper appeared in the *Review of Economics and Statistics* in 2021 (Michaillat and Saez 2021b). In chapter 13 we use the approach developed in the REStat paper to analyze forward guidance in the slackish business cycle model. We find that thanks to the wealth-in-the-utility assumption, the effects of forward guidance are reasonable, and in fact dissipate as the zero-lower-bound episode lasts an increasingly long time.

How large should stimulus packages be?

When monetary policy is constrained by the zero lower bound—as during the Great Depression and the Great Recession—the Federal Reserve is not able to stabilize the unemployment rate to its efficient level. In that case, monetary policy is ineffective, but government spending can be used to stabilize unemployment. This is why the US government designed a large stimulus package during the Great Recession. But it was unclear at the time how large that stimulus package should be, or even if a stimulus package was desirable at all, because there was no framework to answer such a question.

The policy debate motivated Emmanuel and I to study optimal stimulus spending when there is unemployment, and in particular unemployment is inefficiently high. Public

[7] See for instance Zou (1994), Bakshi and Chen (1996), Carroll (2000), Piketty (2011), and Kumhof, Ranciere, and Winant (2015).

economics studied the optimal provision of public goods in normal times, but not in bad times (Samuelson 1954, 1955). Macroeconomics focused on the size of fiscal multipliers in bad times, but not on the optimal amount of government spending in bad times. Using a sufficient-statistic approach again, we established that optimal government spending deviates from the Samuelson rule to reduce, but not eliminate, the unemployment gap. So in a typical slump, in which unemployment is inefficiently high but government spending is able to reduce unemployment, the government should spend above the Samuelson rule—stimulus spending should be positive. The size of the stimulus package depends on several sufficient statistics, including the unemployment gap and the fiscal multiplier. These results were published in the *Review of Economic Studies* in 2019 (Michaillat and Saez 2019), and are covered in chapter 17.

How wide is the unemployment gap?

In the slackish model, unlike in neoclassical models, efficiency is not guaranteed. Since the amount of slack is generally inefficient, policies might help stabilize the economy. As the AEJ and REStud papers show, optimal policies depend critically on whether the current unemployment rate is above or below the socially efficient unemployment rate. The issue is that we did not know what the efficient unemployment rate, u^*, was.

So our next project was to develop a measure of the socially efficient unemployment rate and of the unemployment gap. We found that the socially efficient unemployment rate depended on three sufficient statistics: the cost of recruiting workers, the social value of nonwork, and the elasticity of the Beveridge curve. When we applied the measure to the United States, we found that the US unemployment gap is generally positive and sharply countercyclical. This means that the US labor market is generally inefficient and especially inefficiently slack in slumps. This project was published in the *Journal of Public Economics Plus* in 2021 (Michaillat and Saez 2021a). The results are covered in chapter 9.

In the course of our work on the JPubE paper, Emmanuel and I uncovered that in the United States, servicing a vacancy requires about one worker, home production by jobseekers is virtually nonexistent, and the Beveridge curve is close to a rectangular hyperbola. As a result, our formula for the efficient unemployment rate can be simplified to $u^* = \sqrt{uv}$. The value u^* also measures the full-employment rate of unemployment (FERU), because legal texts define full employment as the amount of employment maximizing social welfare. We also find that since the Great Depression in 1929, the US economy has generally been inefficiently slack, operating short of full employment. These results were published in the *Brookings Papers on Economic Activity* in 2024 (Michaillat and Saez 2024b), and are presented in chapter 12.

Are inflation and slack linked?

A lot of the work in the book was motivated by the significant slack that appeared in the US economy after the Great Recession and during its long recovery. The recovery of the pandemic recession of 2020, by contrast, was fast and led to significant tightness in the US labor market. In fact the US labor market had not experienced such tight conditions since the end of World War 2. These tight conditions led to new questions.

The first question was whether the inefficient tightness experienced on the US labor market could explain the high inflation experienced at the same time, and if so through which mechanism. Until then Emmanuel and I had assumed a fixed-inflation norm because inflation had remained exceedingly stable in the thirty years prior to the pandemic. But the inflation spike after the pandemic pushed us to relax this assumption and study how slack and inflation could be connected.

We therefore developed a new theory of the Phillips curve that connects the unemployment gap to inflation. The Phillips curve emerged from directed search (Moen 1997), combined with a price-adjustment cost (Rotemberg 1982). Because search is directed, workers search for buyers offering the highest prices for their services, knowing that they will face more competition for high-paying jobs and less for low-paying jobs. The price-adjustment cost was required because without it, the Phillips curve would be degenerate: vertical instead of downward-sloping, maintaining unemployment at its efficient level.

The Phillips curve that we developed ensures what Blanchard and Gali (2007) call the divine coincidence: inflation is on target when the economy is at full employment. Consequently, inflation is above target when the economy is inefficiently tight, and below target when the economy is inefficiently slack—in line with the evidence unearthed by Benigno and Eggertsson (2023). This work was circulated on arXiv in 2024 (Michaillat and Saez 2024a) and is the basis for chapter 14.

Does immigration affect slack?

A second question that arose during the post-pandemic overheating is whether there were other policies besides cutting aggregate demand that could help. Naturally, another policy that can help stabilize the economy is to expand aggregate supply via immigration. I find that immigration indeed reduces labor market tightness, and that the effect of immigration on tightness and unemployment depends on the state of the labor market. In particular, when the labor market becomes slacker, local workers are more negatively impacted by immigration, as the competition for jobs is fiercer. I also argue that while immigration has a negative effect on local workers because it increases their unemployment rate, it has a positive effect on local firms because it makes it easier for them to recruit—explaining why immigration is such a polarizing topic. This work on migration was circulated on arXiv in 2023 (Michaillat 2024) and is discussed in chapter 11.

Has a new recession started?

Finally, after labor market tightness peaked in 2022, the US labor market cooled fairly rapidly in 2023–2025. This naturally led to the question: Has a new recession started? The JPubE and BPEA papers show that the combination of vacancy and unemployment data is useful to compute the FERU and unemployment gap in real time. But it turns out that these data are very useful to detect recessions in real time too: recessions feature not only an increase in unemployment but also a decline in job vacancy as the economy moves down the Beveridge curve. Therefore, by combining data on unemployment and job vacancies, we construct a recession rule (Michez rule) that detects recessions faster and more robustly than the popular Sahm (2019) rule. It's even possible to do better than the Sahm and Michez rules by optimizing how the recession rules are constructed—by optimally filtering labor market data and selecting the detection threshold. The Michez rule was published in the *Oxford Bulletin of Economics and Statistics* in 2025 (Michaillat and Saez 2025); the optimal rules were described on arXiv in 2025 (Michaillat 2025); both papers form the basis for the results in chapter 18.

What is new in this book?

Readers familiar with the papers described above will recognize elements of the slackish models and policy analyses presented in the book. However, the book aims to be more than a collection of papers. It provides a unified and systematic treatment of the theory of slack, with consistent notation, concepts, and methodology. It also provides substantial evidence about slack—coming from diverse sources and spanning nearly one hundred years—to motivate and discipline the theory. Finally, it places the theory in context by clarifying its similarities and differences with the main business cycle paradigms—Old Keynesian, Real Business Cycle, and New Keynesian.

The ingredients of the theory presented in this book are not new: the matching function comes from the DMP literature; price and wage rigidities are ubiquitous in the Keynesian literature; wealth in the utility function has appeared before; sufficient statistics are commonplace in public economics; and so on. The novelty lies in how the book curates and combines these ingredients to offer a slack-centered theory of how economies operate and fluctuate, and to provide slack-based tools to monitor and respond to the cyclical inefficiencies that markets inevitably generate.

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